Year 11 Higher

REVISION PACK

Graphs 6

Name: _____

Class: _____

Graphs 6 · Graphs to solve equations (Lessons 8.1, 8.2)

Worked example: solving with a quadratic graph

The graph of $y = x^2 - 2x - 2$ is drawn on a grid.

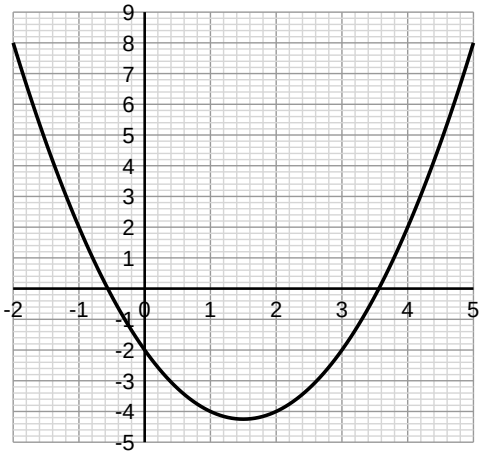
(a) Use the graph to find estimates for the solutions of $x^2 - 2x - 2 = 0$

(b) By drawing a suitable straight line, find estimates for the solutions of $x^2 - 3x - 1 = 0$

(a)	**Step 1**	Read where the curve crosses the x-axis	
	Step 2	Answer	$x = -0.7$ or $x = 2.7$
(b)	**Step 1**	Make the left side the curve	$x^2 - 2x - 2 = x - 1$
	Step 2	Draw the line	$y = x - 1$
	Step 3	Read where they cross	
	Step 4	Answer	$x = -0.3$ or $x = 3.3$

Practice: solving with a quadratic graph

1.1

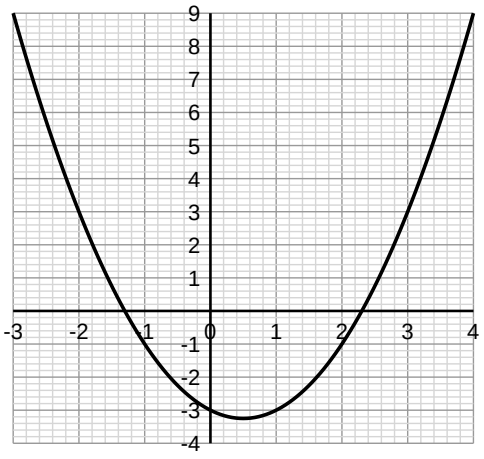


The graph of $y = x^2 - 3x - 2$ is drawn on the grid.

(a) Use the graph to find estimates for the solutions of $x^2 - 3x - 2 = 0$

(b) By drawing a suitable straight line on the grid, find estimates for the solutions of $x^2 - 2x - 4 = 0$

1.2



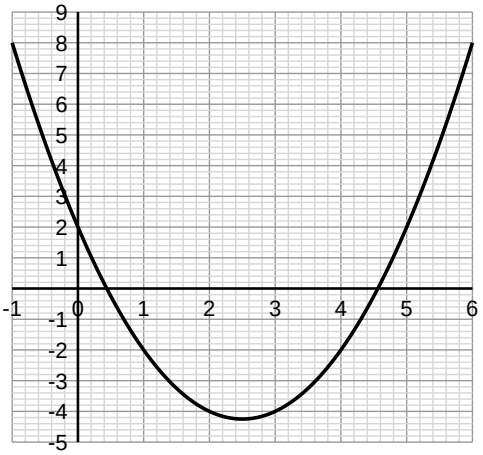
The graph of $y = x^2 - x - 3$ is drawn on the grid.

(a) Use the graph to find estimates for the solutions of $x^2 - x - 3 = 0$

(b) By drawing a suitable straight line on the grid, find estimates for the solutions of $x^2 - 2x - 1 = 0$

Practice: solving with a quadratic graph (continued)

1.3

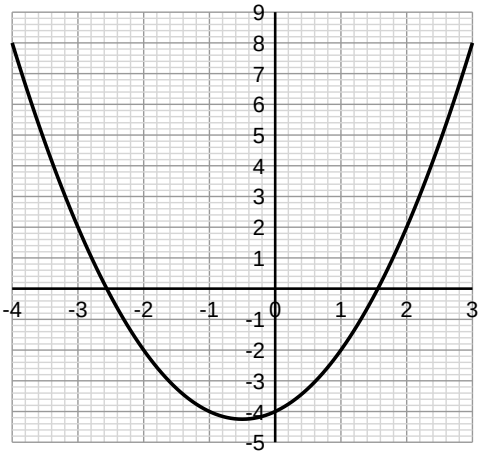


The graph of $y = x^2 - 5x + 2$ is drawn on the grid.

(a) Use the graph to find estimates for the solutions of $x^2 - 5x + 2 = 0$

(b) By drawing a suitable straight line on the grid, find estimates for the solutions of $x^2 - 4x - 1 = 0$

1.4



The graph of $y = x^2 + x - 4$ is drawn on the grid.

(a) Use the graph to find estimates for the solutions of $x^2 + x - 4 = 0$

(b) By drawing a suitable straight line on the grid, find estimates for the solutions of $x^2 - 2 = 0$

Graphs 6 · Graphs to solve equations (Lessons 8.1, 8.2)

Worked example: solving with a cubic graph

The graph of $y = x^3 - 3x + 2$ is drawn on a grid.

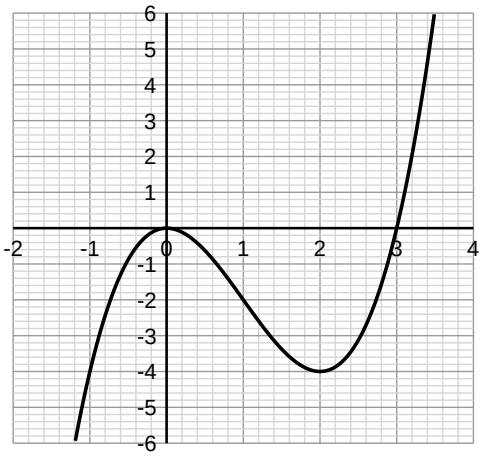
(a) Find estimates for the solutions of $x^3 - 3x + 2 = 1$

(b) The equation $x^3 - 3x + 2 = k$ has three solutions. Write down the range of values of k.

(a)	Step 1	Draw the line	$y = 1$
	Step 2	Read where they cross	
	Step 3	Answer	$x = -1.9,\ 0.3,\ 1.5$
(b)	Step 1	Read the turning points	$y = 4$ and $y = 0$
	Step 2	Answer	$0 < k < 4$

Practice: solving with a cubic graph

1.5

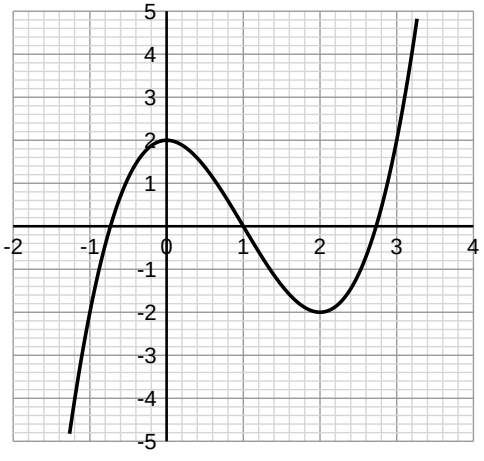


The graph of $y = x^3 - 3x^2$ is drawn on the grid.

(a) Use the graph to find estimates for the solutions of $x^3 - 3x^2 = -2$

(b) The equation $x^3 - 3x^2 = k$ has three solutions. Write down the range of possible values of k.

1.6



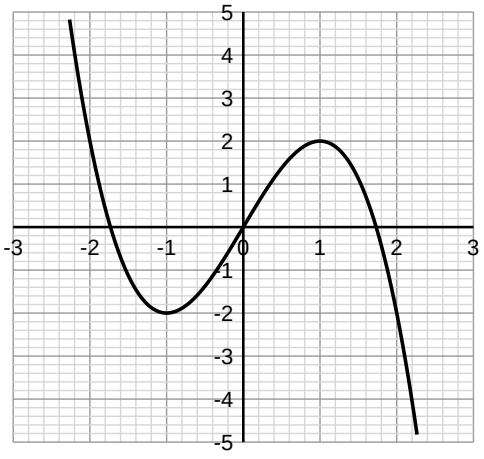
The graph of $y = x^3 - 3x^2 + 2$ is drawn on the grid.

(a) Use the graph to find estimates for the solutions of $x^3 - 3x^2 + 2 = 1$

(b) The equation $x^3 - 3x^2 + 2 = k$ has three solutions. Write down the range of possible values of k.

Practice: solving with a cubic graph (continued)

1.7

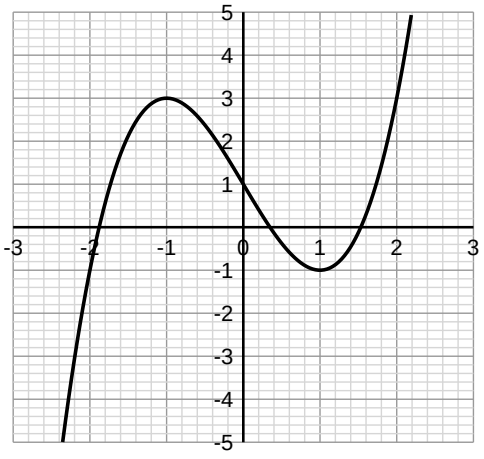


The graph of $y = 3x - x^3$ is drawn on the grid.

(a) Use the graph to find estimates for the solutions of $3x - x^3 = -1$

(b) The equation $3x - x^3 = k$ has three solutions. Write down the range of possible values of k.

1.8



The graph of $y = x^3 - 3x + 1$ is drawn on the grid.

(a) Use the graph to find estimates for the solutions of $x^3 - 3x + 1 = 1$

(b) The equation $x^3 - 3x + 1 = k$ has three solutions. Write down the range of possible values of k.

Graphs 6 · Non-linear simultaneous equations (Lesson 8.3)

Worked example: a line and a circle

Solve the simultaneous equations $y = x + 7$ and $x^2 + y^2 = 25$

Step 1	Substitute	$x^2 + (x+7)^2 = 25$
Step 2	Expand	$x^2 + x^2 + 14x + 49 = 25$
Step 3	Collect, subtract 25	$2x^2 + 14x + 24 = 0$
Step 4	Divide by 2	$x^2 + 7x + 12 = 0$
Step 5	Factorise	$(x+3)(x+4) = 0$
Step 6	Find y	$y = -3 + 7 = 4, \quad y = -4 + 7 = 3$
Step 7	Answer	$x = -3, y = 4$ or $x = -4, y = 3$

Practice: a line and a circle

2.1 Solve the simultaneous equations

$y = x - 1$

$x^2 + y^2 = 13$

2.2 Solve the simultaneous equations

$y = 2x - 5$

$x^2 + y^2 = 10$

2.3 Solve the simultaneous equations

$y = x + 3$

$x^2 + y^2 = 17$

2.4 Solve the simultaneous equations

$y = -x + 5$

$x^2 + y^2 = 13$

2.5 Solve the simultaneous equations

$y = 3x - 10$

$x^2 + y^2 = 20$

2.6 Solve the simultaneous equations

$y = x - 2$

$x^2 + y^2 = 10$

Graphs 6 · Non-linear simultaneous equations (Lesson 8.3)

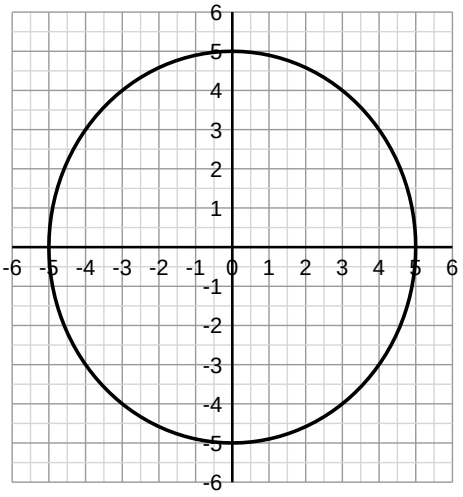
Worked example: a line and a circle on a graph

The graph of $x^2 + y^2 = 25$ is drawn on a grid. By drawing a suitable straight line, solve $x^2 + y^2 = 25$ and $y = x - 7$

Step 1	Draw the line	$y = x - 7$
Step 2	Read where they cross	$(3, -4)$ and $(4, -3)$
Step 3	Answer	$x = 3, y = -4$ or $x = 4, y = -3$

Practice: a line and a circle on a graph

2.7



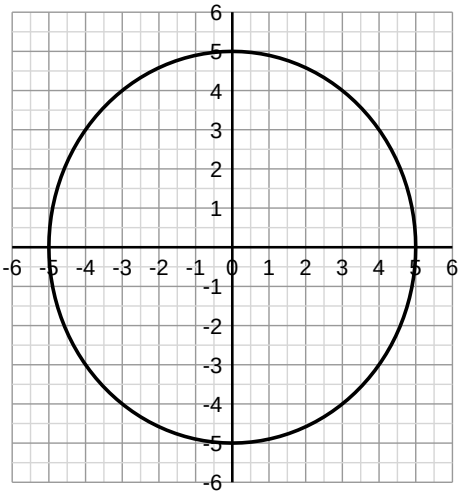
The graph of $x^2 + y^2 = 25$ is drawn on the grid.

By drawing a suitable straight line on the grid, find the solutions of the simultaneous equations

$x^2 + y^2 = 25$

$y = x - 1$

2.8



The graph of $x^2 + y^2 = 25$ is drawn on the grid.

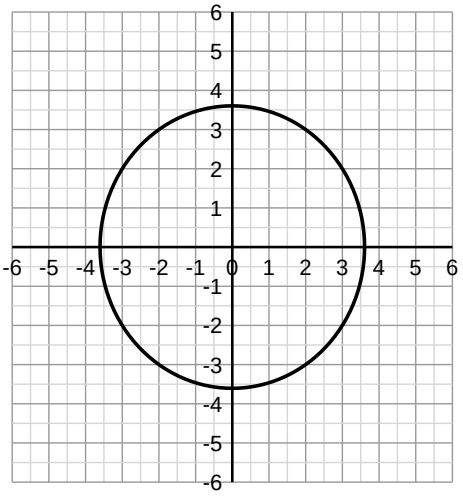
By drawing a suitable straight line on the grid, find the solutions of the simultaneous equations

$x^2 + y^2 = 25$

$y = -x + 7$

Practice: a line and a circle on a graph (continued)

2.9



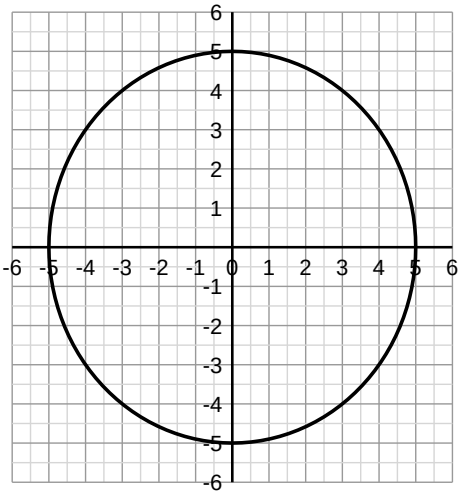
The graph of $x^2 + y^2 = 13$ is drawn on the grid.

By drawing a suitable straight line on the grid, find the solutions of the simultaneous equations

$x^2 + y^2 = 13$

$y = x + 1$

2.10



The graph of $x^2 + y^2 = 25$ is drawn on the grid.

By drawing a suitable straight line on the grid, find the solutions of the simultaneous equations

$x^2 + y^2 = 25$

$y = -x + 1$

Graphs 6 · Non-linear simultaneous equations (Lesson 8.3)

Worked example: a line and a quadratic

Solve the simultaneous equations $y = 2x + 1$ and $y = x^2 - 2$

Step 1	Make them equal	$x^2 - 2 = 2x + 1$
Step 2	Make one side 0	$x^2 - 2x - 3 = 0$
Step 3	Factorise	$(x - 3)(x + 1) = 0$
Step 4	Each bracket = 0	$x = 3$ or $x = -1$
Step 5	Find y	$y = 2 \times 3 + 1 = 7, \quad y = 2 \times (-1) + 1 = -1$
Step 6	Answer	$x = 3, y = 7$ or $x = -1, y = -1$

Practice: a line and a quadratic

3.1 Solve the simultaneous equations

$y = x + 5$

$y = x^2 - 1$

3.2 Solve the simultaneous equations

$y = 2x + 3$

$y = x^2$

3.3 Solve the simultaneous equations

$y = 3x - 2$

$y = x^2 - 2x + 4$

3.4 Solve the simultaneous equations

$y = -x + 4$

$y = x^2 - 2$

3.5 Solve the simultaneous equations

$y = 2x - 1$

$y = x^2 + x - 3$

3.6 Solve the simultaneous equations

$y = 4x - 3$

$y = x^2$